Name: ID#:

Problem 1 Bayes Classifiers, *(10 points.)*

x_1	x_2	x_3	y
0	1	1	0
0	1	1	0
1	0	1	0
1	1	0	0
0	0	1	1
0	1	0	1
1	0	0	1
1	0	0	1

Consider the table of measured data given at right. We will use the three observed features x_1, x_2, x_3 to predict the class y. In the case of a tie, we will prefer to predict class $y = 0$.

(1) Write down the probabilities used by a naïve Bayes classifier: *(4 points.)*

$p(y=0)$: $\frac{1}{2}$ $p(y=1)$: $\frac{1}{2}$

$p(x_1=1|y=0)$: $\frac{1}{2}$ $p(x_1=1|y=1)$: $\frac{1}{2}$

$p(x_1=0|y=0)$: $\frac{1}{2}$ $p(x_1=0|y=1)$: $\frac{1}{2}$

$p(x_2=1|y=0)$: $\frac{3}{4}$ $p(x_2=1|y=1)$: $\frac{1}{4}$

$p(x_2=0|y=0)$: $\frac{1}{4}$ $p(x_2=0|y=1)$: $3/4$

$p(x_3=1|y=0)$: $3/4$ $p(x_3=1|y=1)$: $1/4$

$p(x_3=0|y=0)$: $1/4$ $p(x_3=0|y=1)$: $3/4$

(2) Using your naïve Bayes model, compute: *(3 points.)*

$p(y=0|x_1=0,x_2=0,x_3=0)$:

$$\frac{\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{4}\cdot\frac{1}{4}}{\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{4}\cdot\frac{1}{4}+\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{3}{4}\cdot\frac{3}{4}}$$

$= 0.1$

$p(y=1|x_1=0,x_2=0,x_3=0)$:

$1 - 0.1 = 0.9$

(3) Compute the probabilities $p(x_1=0,x_2=0,x_3=0|y=0)$ and $p(x_1=0,x_2=0,x_3=0|y=1)$ for a joint Bayes model trained on the same data. *(3 points.)*

$P(x_1=0,x_2=0,x_3=0 \mid y=0) = 0$

$P(x_1=0,x_2=0,x_3=0 \mid y=1) = 0$

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Problem 2 Decision Trees, *(10 points.)*

Consider the table of measured data given at right. (Note that some data points are repeated.) We will use a decision tree to predict the outcome y using two features, x_1, x_2, where each can take one of three values: a, b, c. In the case of ties, we prefer to use the feature with the smaller index (x_1 over x_2, etc.) and prefer to predict class 1 over class 0. You may find the following values useful (although you may also leave logs unexpanded):

x_1	x_2	y
a	a	1
a	b	0
a	c	0
a	a	1
b	c	1
b	a	1
c	c	1
c	a	0
c	b	0
c	c	1

$\log_2(1) = 0$ $\log_2(2) = 1$ $\log_2(3) = 1.59$ $\log_2(4) = 2$
$\log_2(5) = 2.32$ $\log_2(6) = 2.59$ $\log_2(7) = 2.81$ $\log_2(8) = 3$

(1) What is the entropy of y? *(2 points.)*

$P(y=1) = 0.6$ $H(y) = -0.6 \log 0.6 - 0.4 \log 0.4$

0.966 or 0.971 ← calculator

(2) What is the information gain of x_1 and x_2? *(4 points.)*

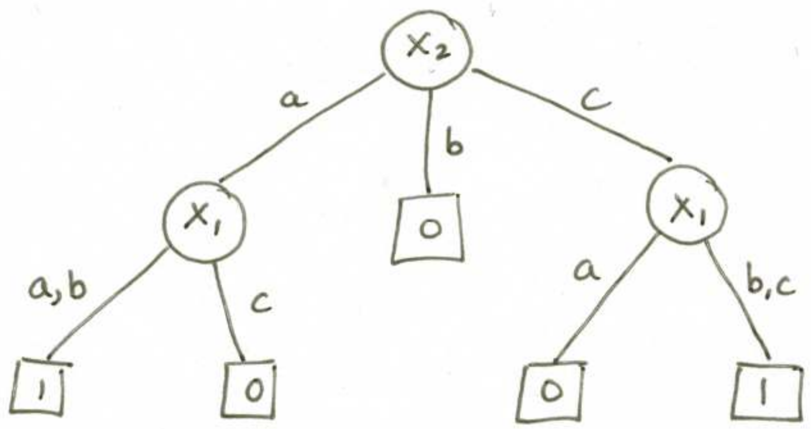
$IG(x_1)$: $H(y) - \frac{2}{5}H(\frac{1}{2}) - \frac{1}{5}H(0) - \frac{2}{5}H(\frac{1}{2}) = H(y) - \frac{4}{5} \sim 0.166$

$IG(x_2)$: $H(y) - \frac{2}{5}H(\frac{1}{4}) - \frac{1}{5}H(0) - \frac{2}{5}H(\frac{1}{4}) = H(y) - \frac{4}{5}H(\frac{1}{4}) \sim 0.320$

$H(\frac{1}{4}) = 2 - \frac{3}{4}\log 3 \sim 0.808$

$IG(x_2) > IG(x_1)$

(3) Based on the information gain computed in (2), build the complete decision tree learned on this data. *(4 points.)*



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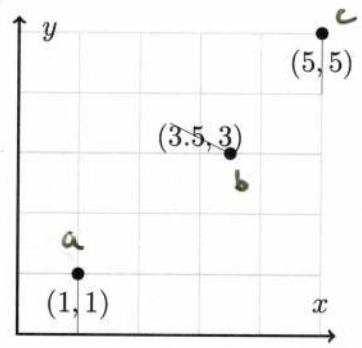
Problem 3 Linear and Nearest Neighbor Regression, *(8 points.)*

Consider the data points shown at right, for a regression problem to predict y given a scalar feature x.



(1) Compute **training** MSE of a 1-nearest neighbor predictor. *(2 points.)*

0

(2) Compute the **leave-one-out** cross-validation error (MSE) of a 1-nearest neighbor predictor. *(3 points.)*

	pred, $\hat{y}$	true, y
a)	3	1
b)	5	3
c)	3	5

$$MSE = \frac{1}{3}\left((3-1)^2 + (5-3)^2 + (3-5)^2\right)$$

$$= 4$$

(3) Compute the **leave-one-out** cross-validation error (MSE) of a linear regressor. *(3 points.)*

	Pred, $\hat{y}$	true, y
a)	$-\frac{1}{3}$	1
b)	3.5	3
c)	4.2	5

$$MSE = \frac{1}{3}\left(\left(-\frac{1}{3}-1\right)^2 + (3.5-3)^2 + (4.2-5)^2\right)$$

$$= \frac{1}{3}\left(\frac{16}{9} + \frac{1}{4} + \frac{16}{25}\right)$$

$$\sim 0.889$$

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Problem 4 Multiple Choice, *(10 points.)*

Here, assume that we have m data points $y^{(i)}$, $x^{(i)}$, $i = 1 \ldots m$, each with n features, $x^{(i)} = [x_1^{(i)} \ \ldots \ x_n^{(i)}]$. For each of the choices below, will it likely increase, decrease, or have no effect on overfitting (circle your choice)? If you think it is equally likely to go either way, pick *No Effect*.

1	Gathering more labeled training data	**Reduce**	**Increase**	**No Effect**
2	Create $4 \times m$ training data by copying it	**Reduce**	**Increase**	**No Effect**
3	Increasing K for KNN Classifiers	**Reduce**	**Increase**	**No Effect**
4	Rescaling the training data (shift+scale)	**Reduce**	**Increase**	**No Effect**
5	Adding features that are just random values	**Reduce**	**Increase**	**No Effect**
6	Adding another layer to a Neural Network	**Reduce**	**Increase**	**No Effect**
7	Changing the activation function of hidden nodes	**Reduce**	**Increase**	**No Effect**
8	Switching from soft to hard margin SVMs	**Reduce**	**Increase**	**No Effect**
9	Switching from linear to polynomial Kernel SVMs	**Reduce**	**Increase**	**No Effect**
10	Increasing `MinParent` of a Decision Tree	**Reduce**	**Increase**	**No Effect**